

AI-POWERED PREDICTIVE TELEMETRY & INDUSTRIAL DIAGNOSTICS

# MechMind AI

Intelligent Condition Monitoring & Generative Diagnostics for Heavy Machinery

Connecting industrial IoT edge telemetry with technicians and fleet managers via WhatsApp

## 75% Diagnostic Accuracy

### 56-Question Blind Benchmark

Validated across hydraulic, electrical, engine & powertrain faults

## Under \$25 Hardware BOM

### ESP32 + Tri-Axial Sensors

Replaces \$10,000+ imported proprietary diagnostic tools

## 90-Second WhatsApp Triage

### Real-Time Telematics & RAG

Delivers verified OEM repair procedures & part numbers

## \$500 — \$2,500 Lost Daily Per Stalled Machine

Unplanned mechanical downtime in construction and agriculture creates severe project delays, idle labour, and wasted repair costs.

### 1. Closed OEM Systems & Language Barrier

Heavy equipment brands (Caterpillar, Komatsu, XCMG, SANY) lock telemetry behind closed protocols. Manuals and fault codes are technical and often in foreign languages that local mechanics cannot decipher.

### 2. Trial-and-Error Misdiagnosis

With no affordable diagnostics available, technicians guess faults by trial and error, replacing expensive functional parts blindly. This causes recurring breakdowns, safety hazards, and weeks of idle downtime waiting for imported parts.

### 3. Exorbitant Imported Scanners

Foreign OEM diagnostic tools and cloud subscriptions cost \$5,000 to \$10,000+ per brand. Small and medium contractors, plant hire firms, and agricultural cooperatives are completely priced out.

Layer 1: The MechMind Node (IoT Edge Hardware)

**Magnetic Clamping:** Installs in 30 seconds on any machine surface without drilling or voiding warranties.

**Multi-Modal Sensing:** Tri-axial vibration (MPU-6050), surface temperature (DS18B20 probe), acoustic microphor

**Edge Intelligence:** Dual-core ESP32 calculates real-time RMS vibration and flags anomalies using ISO 10816.

**Ultra-Low Cost:** Bills at under \$25 (NGN 35,000) in components vs \$10,000+ imported proprietary units.

**Rechargeable Power:** Integrated 18650 Li-ion battery + pass-through charging for 12+ hours continuous field op

Layer 2: WhatsApp AI Assistant & Fleet Dashboard

**Zero Friction Access:** No apps to download. Mechanics interact directly through WhatsApp (+234 701 029 9562).

**Multi-Modal Inputs:** Accepts typed fault codes, voice notes (Whisper speech-to-text), and broken component photos

**RAG Grounded Intelligence:** Responses strictly verified against indexed OEM manuals (XCMG, SANY, Caterpillar

**Actionable Output:** Delivers probable root cause, step-by-step repair guide, torque limits, and OEM parts in 90s.

**Fleet Cloud Dashboard:** Web console (mechmind-ng.vercel.app) gives fleet managers real-time telemetry waveform

75.0%

Diagnostic Accuracy

Tested on 56-question blind test suite across hydraulics, electrical & powertrain

< 90 Sec

Average Triage Speed

Cuts diagnostic downtime from 2+ days waiting for specialists down to minutes

Live & Public

WhatsApp Assistant

Active WhatsApp bot (+234 701 029 9562) handling multi-turn diagnostics

Deployed Cloud

Fleet Telemetry Hub

Production dashboard live at mechemind-ng.vercel.app with real-time streaming

Real-World Field Validation & Technical Readiness

- 1. **Physical Hardware Assembled & Tested:** Sensor nodes assembled with ESP32, MPU-6050 accelerometer, and DS18B20 temperature probes, encased in rugged magnetic shells.
- 2. **Frontline Workshop & Field Validation:** Grounded in hands-on SIWES industrial experience across civil construction, transport corridors, and heavy equipment workshops.
- 3. **Full Software & RAG Pipeline Deployed:** High-throughput FastAPI backend, ChromaDB vector indexing of OEM service manuals, quantized Llama 3.1 8B LLM, and live dashboard.
- 4. **Commercial Entity Registered:** Pullup Events (CAC Registration BN-9846744) covering Technology & Software services.

\$45 Billion+

African Heavy Equipment Market

Rapid infrastructure expansion, mining growth, and agricultural mechanization

350,000+

Active Machines in Region

Excavators, bulldozers, tractors, wheel loaders, and commercial generators

45,000+

Contractors & Fleet Hubs

Civil contractors, plant hire operators, commercial farms, and quarry sites

Target Customer Segments & Deployment Strategy

Civil Engineering Contractors

Mid-sized contractors with 5–30 machines who cannot afford full-time diagnostic specialists. MechMind prevents costly job-site penalties.

Heavy Equipment Plant Hire Firms

Rental operators who lose revenue when machines break down on client sites. Real-time telemetry protects rental assets from operator abuse.

Agricultural Fleets & Mechanization Hubs

Tractor cooperatives and irrigation sites where equipment downtime directly threatens crop planting, harvesting, and food security.

1. Hardware Node Sales

\$50 – \$75 per node

One-Time Purchase (45% Margin)

Rugged ESP32 sensor node sold with healthy gross margin over \$25 BOM. Magnetic clip-on design makes self-installation effortless for fleet operators.

2. SaaS Telemetry Subscription

\$15 – \$25 / machine

Monthly Recurring (MRR)

Continuous live telemetry monitoring, automated ISO 10816 anomaly alerts, fleet health analytics dashboard, and unlimited priority WhatsApp AI queries.

3. Spare Parts & Service Commission

3% – 5% per order

Marketplace Transaction Fee

Direct matchmaking with certified OEM parts suppliers. Mechanics order verified replacement parts directly via the WhatsApp diagnostic flow.

Mubarak Raji Babatunde

Founder & AI / Software Lead

Software & Artificial Intelligence Engineer

Full-stack and AI developer. Engineered the MechMind RAG pipeline, WhatsApp bot, FastAPI ingestion backend, and Next.js cloud dashboard. Experienced in low-latency API architectures.

Olawale Oladele

Hardware Design & Embedded Systems Lead

Mechanical & Embedded Systems Engineer

CSWP SolidWorks Professional certified with 4 CSWPA specializations and Ansys FEA credentials. Designed the 3D-printed enclosure, sensor circuit schematics, and FreeRTOS firmware algorithms.

Oni Mujeeb Ramadan

Product Operations & Field Validation Lead

Operations & Technical Validation Lead

NUESA Engineering Scholar of the Year (2026), Man of Excellence Award (LASUSU). Manages site field tests, equipment operator interviews, diagnostic manual collation, and partner outreach.

Phase 1 (Months 1 – 3)

Pilot Production & Lab Benchmarking

Target: 50 Pilot Units

- Manufacture first 50 production-grade SMT PCB sensor nodes.
- Expand manual indexing to 20 Caterpillar, Komatsu & SANY models.
- Deploy pilot units across 10 commercial fleets.
- Achieve 80%+ diagnostic accuracy on expanded test set.

Phase 2 (Months 4 – 6)

Fleet Deployment & Cellular Telemetry

Target: 25 Commercial Fleets

- Onboard 25 commercial civil engineering contractors and plant hire fleets.
- Implement automated GSM/cellular telemetry fallback for off-grid sites.
- Integrate direct spare part ordering through WhatsApp.
- Target sustainable unit economics and recurring MRR.

Phase 3 (Months 7 – 9)

Multi-Brand Expansion & Damage Vision

Target: 50+ Models & Vision AI

- Index 35+ equipment brands (Komatsu, Shantui, SANY, Hitachi, Perkins).
- Deploy localized multimodal vision model for heavy machinery wear detection.
- Expand field operations into regional transport and mining corridors.
- Secure distribution partnerships with major equipment dealerships.

Phase 4 (Months 10 – 12)

Pan-African Commercial Scaling

Target: 500+ Active Nodes

- Scale to 500+ active hardware sensor nodes across 75+ fleets.
- Launch predictive remaining useful life (RUL) enterprise dashboard.
- Expand across West and East Africa under AfCFTA Right-to-Repair framework.
- Position MechMind AI as Africa's universal equipment telemetry standard.

# Democratizing Industrial Diagnostics Across Africa

MechMind AI — Transforming Maintenance from Reactive Failure to Predictive Intelligence

## Verify MechMind AI Live Today

- **WhatsApp Diagnostic Bot:** [+234 701 029 9562](https://wa.me/2347010299562)
- **Live Telemetry Dashboard:** <https://mechmind-ng.vercel.app>
- **Open-Source Codebase:** [github.com/DevMubarak1/mechmind-ai](https://github.com/DevMubarak1/mechmind-ai)
- **CAC Business Registration:** [Pullup Events \(CAC BN-9846744\)](#)